Measurement Notes — 1. Errors

A measurement reports a value together with a unit.

Random error changes unpredictably between repeated measurements.

Systematic error shifts results in a consistent direction.

Repeating and averaging can reduce random variation.

Averaging does not by itself remove a systematic offset.

Example: a balance that reads 0.20 g too high has an offset.

Measurement Notes — 2. Reporting

Three measured masses: 10.1 g, 10.2 g, 10.3 g.

Their arithmetic mean is 10.2 g.

Report the unit and how the value was obtained.

This handout does not provide instrument uncertainty.

Do not infer instrument uncertainty from the displayed decimal places.

The example balance offset is not stated to apply to these three masses.